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ARIA-NBV: Target-Conditioned, Quality-Driven Next-Best-View Planning

ARIA-NBV: Zielkonditionierte, qualitaetsgetriebene Next-Best-View-Planung

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Transparency in the Use of AI Tools

AI-assisted tools were used to organize literature notes, check consistency across repository documentation, and draft parts of the proposal text. The author remains responsible for the final research scope, technical claims, citations, and submitted document.

Contents

1 Motivation

Active reconstruction is a coupled sensing and inference problem: the reconstruction error after a fixed budget depends as much on the selected viewpoints as on the downstream surface estimator. Classical active perception therefore treats sensing actions as part of perception itself, and view-planning surveys formalize the dominant generate-score-select loop for three-dimensional inspection [AWB88, Baj88, SRR03]. ARIA-NBV adopts that loop for egocentric indoor data, but asks a narrower question: when views are restricted to a finite feasible candidate table, does reconstruction-quality improvement contain planning structure beyond one-step selection?

The key empirical precedent is VIN-NBV, which replaces pure coverage with Relative Reconstruction Improvement (RRI), an oracle label computed from point-mesh reconstruction-error reduction after adding a query view [Fra+25]. ARIA-NBV uses the same quality-driven axis because target indoor surfaces can remain poor even when coverage proxies look saturated. The thesis transfers this idea to the Project Aria / ASE regime, where calibrated egocentric streams, trajectories, semi-dense points, predicted object boxes, and mesh-supervised assets support controlled oracle labels [Eng+23, Met25, Str+24a].

The implemented seminar substrate already provides scene-level oracle RRI labels and a VINv3-style one-step candidate scorer on frozen EVL features. This proposal treats that as the starting point, not the final thesis result. The extension is target-specific: actor-visible target selection, target-cropped oracle labels, replayable counterfactual rollouts, and a finite-horizon value model over candidate rows.

Continuous and hierarchical NBV papers motivate later directions, but they do not define the first thesis test. GenNBV and Hestia assume mature simulator dynamics and reward loops [Che+24, Lu+26]; active NeRF and 3DGS work motivates utility-channel diagnostics rather than replacing the mesh-supervised RRI objective [BLL25, Jeo+26, JLD24, LJD25, Pan+22, Str+24b].

Thesis position

ARIA-NBV tests target-conditioned, quality-driven NBV on ASE/EFM as a finite-candidate planning problem. The core experiment first measures whether bounded oracle lookahead exposes non-myopic target-RRI headroom. If it does, a masked candidate-query Q_H model is evaluated by how much of that headroom it recovers from actor-visible rollout data.

2 Problem and Research Contract

ARIA-NBV is a target-conditioned, finite-candidate NBV problem in the Project Aria / ASE / EFM observation regime. The thesis is deliberately not a first-order continuous-control claim: it tests whether bounded planning over a finite valid candidate table improves target reconstruction quality beyond myopic view selection.

The actor-visible state at decision step t is

$$\boldsymbol{s}_t^{\text{obs}} = \left(\boldsymbol{P}_t^{\text{semi}}, \boldsymbol{F}_t^{\text{EVL}}, \boldsymbol{O}_t^{\text{pred}}, \boldsymbol{h}_t, b_t\right),$$

where $\boldsymbol{P}_t^{\text{semi}}$ is the accumulated semi-dense or fused point evidence, $\boldsymbol{F}_t^{\text{EVL}}$ is the frozen local EFM/EVL evidence field, $\boldsymbol{O}_t^{\text{pred}}$ are observed or predicted target hypotheses, $\boldsymbol{h}_t$ is selected-view history, and b_t is remaining budget. The oracle state augments this with ASE GT assets:

$$\boldsymbol{s}_t^{\text{oracle}} = \left(\boldsymbol{s}_t^{\text{obs}}, \boldsymbol{M}_{\text{GT}}, \left\{\boldsymbol{M}_e^{\text{GT}}\right\}_{e\in\mathcal{E}}, \left\{\boldsymbol{P}_{t,i}^{\text{cand}}\right\}_{i=1}^{N_q}\right).$$

GT meshes, GT OBBs, GT crops, GT semantic labels, and all-candidate GT renders are label/evaluation assets only. They are not V1 actor inputs.

The planner state is

$$\boldsymbol{s}_t^{\text{cf0}} = (\boldsymbol{F}_0^{\text{EVL}}, \boldsymbol{P}_t, \boldsymbol{z}_e, \boldsymbol{h}_t, b_t, \boldsymbol{Q}_t, \boldsymbol{m}_t, \boldsymbol{\rho}_t),$$

where $\boldsymbol{Q}_t = \{\boldsymbol{q}_{t,i}\}_{i=1}^{N_q}$ is the finite candidate table, $m_{t,i} \in \{0,1\}$ is a hard validity mask, and $\rho_{t,i}$ is an invalid reason. The admissible actions are candidate indices:

$$\mathcal{A}_t = \left\{i \in \{1, ..., N_q\} : m_{t,i} = 1\right\}.$$

Invalidity is a constraint, not low utility. Masks apply before argmax, softmax, loss targets, and bootstrap maximization.

Counterfactual rollouts keep the root EVL field $\boldsymbol{F}_0^{\text{EVL}}$ fixed. After a selected valid view, only the accumulated geometry proxy, history, budget, candidate table, masks, and reason codes are updated. The proposal therefore studies geometry-only counterfactual planning, not synthetic future EVL re-inference.

The V1 target protocol is OBS-SEL / PRED-Q / GT-EVAL. Target selection and model input use an actor-visible descriptor

$$\boldsymbol{z}_e = \varphi\Big(\hat{\boldsymbol{B}}_e, \hat{\boldsymbol{y}}_e, \hat{p}_e, A_e^{\text{proj}}, n_e^{\text{semi}}, n_e^{\text{EVL}}, \boldsymbol{T}_e^{\text{rel}}\Big),$$

covering predicted/observed OBB geometry, class, confidence, projected area, semi-dense support, EVL support, and relative pose. GT crops $\boldsymbol{M}_e$ are used only after matching $\boldsymbol{z}_e$ to GT. The matching score is protocol-level, not an actor input:

$$\mu(\hat{e}, e) = \kappa(\hat{y}_{\hat{e}}, y_e) \cdot \mathrm{IoU}_{3\mathrm{D}\left(\hat{\boldsymbol{B}}_{\hat{e}}, \boldsymbol{B}_e^{\mathrm{GT}}\right)} \cdot \sigma\Big(A_{\hat{e}}^{\text{proj}}, n_{\hat{e}}^{\text{semi}}, n_{\hat{e}}^{\text{EVL}}\Big).$$

Unmatched or ambiguous targets are target-invalid cases and are counted separately from low target RRI.

For target e, the implemented oracle error is the point-mesh accuracy plus mesh-to-point completeness diagnostic:

$$\Delta_t^e = \mathcal{A}_t^e + \mathcal{C}_t^e.$$

These are the `pm_acc_*` / `pm_comp_*` point-mesh terms used by `aria_nbv`, not a point-cloud-to-point-cloud distance. If valid action $a_t = i$ selects $\boldsymbol{q}_{t,i}$, then

$$\boldsymbol{P}_{t+1} = \boldsymbol{P}_t \cup \boldsymbol{P}_{t,i}^{\text{cand}}, \quad r_t^e = \frac{\Delta_t^e - \Delta_{t+1}^e}{\Delta_t^e + \varepsilon}.$$

The value-learning return and endpoint metric stay separate:

$$G_t^H = \sum_{k=0}^{H-1} \gamma^k r_{t+k}^e, \quad J_{e,H} = \frac{\Delta_0^e - \Delta_H^e}{\Delta_0^e + \varepsilon}.$$

The log-gain companion $L_{e,H} = \log(\Delta_0^e + \varepsilon) - \log(\Delta_H^e + \varepsilon)$ is only an ablation.

The non-myopic headroom estimate is

$$\Delta_{\text{look}} = J_e^H(\pi_{\text{oracle-look}}) - J_e^H(\pi_{\text{oracle-1}}).$$

If Δ_{look} is approximately zero, the thesis reports that the current candidate distribution and target-RRI objective are effectively myopic. If it is positive, the learned planner is judged by recovery over the one-step actor-visible target scorer:

$$\eta_Q = \frac{J_e^H(\pi_Q) - J_e^H(\pi_{\text{learned-1}})}{J_e^H(\pi_{\text{oracle-look}}) - J_e^H(\pi_{\text{learned-1}}) + \varepsilon}.$$

Main question

Can a target-conditioned candidate-query Transformer $Q_{H,\theta}(\boldsymbol{s}_t^{\text{cf0}}, \boldsymbol{z}_e, \boldsymbol{q}_{t,i})$ recover measurable oracle-lookahead headroom over learned one-step target scoring under equal candidate and acquisition budgets?

The thesis-core evidence is observed target selection, target RRI labels, target-conditioned one-step scoring, replayable oracle rollouts, and masked finite-candidate Q_H. Continuous control, external simulators, 3DGS control, SceneScript, VLM planning, and real-device guidance are follow-up designs unless the finite-candidate result first justifies them.

3 Related Work and Positioning

The literature is used here to assign roles, not to broaden the thesis claim. Older active perception motivates action-conditioned sensing; VIN-NBV supplies the quality-driven candidate-ranking precedent; Project Aria, ASE, and EFM3D define the egocentric state; and offline value learning supplies replay and overestimation controls for the finite candidate table.

Role	Relevant signal	Adopt / defer
Quality-driven NBV [CMR19, Fra+25]	Oracle RRI and ordinal one-step candidate ranking are the closest implemented precedent.	Adopt point-mesh RRI labels and a learned one-step target scorer; test whether one-step ranking is enough.
Egocentric substrate [Eng+23, Met25, Str+24a]	Calibrated egocentric streams, semi-dense points, predicted OBBs, EVL fields, and GT meshes coexist.	Use observed/predicted target descriptors as actor input; keep GT geometry for labels and evaluation.
Greedy sensing and finite candidates [Bir+16, GK11, Jia+25, KSG08, Lau+20]	When utility has diminishing returns, greedy selection can be strong; deeper search must earn its cost empirically.	Measure oracle-lookahead headroom before claiming a learnable non-myopic advantage.
Continuous and radiance-field NBV [Che+24, Jeo+26, JLD24, Lu+26, Pan+22]	Continuous policies, target-then-pose hierarchies, and uncertainty/semantic utility channels are useful comparisons.	Use as follow-up design pressure; do not replace target RRI with coverage or uncertainty rewards.
Finite-action value learning [HGS15, KNL21, Lee+19, Mni+13, Vas+17, Zah+17]	Replay, masked Bellman targets, overestimation control, offline support, and permutation-aware candidate-token modeling shape Q_H.	Train masked fitted Double-Q first; keep IQL, sequence decoding, and continuous actor-critic variants as later ablations.

Table 1: Source-backed literature roles for the proposal scope.

The resulting lineage is deliberately narrow:

$$\mathcal{U}_{\text{cov/unc}} \to \hat{r}_t^e(i) \to r_t^e \to G_t^{(H)} \to Q_{H,\theta}.$$

Coverage and uncertainty remain diagnostics, not the thesis utility. GT meshes and GT target boxes remain oracle assets, not V1 actor inputs. Offline and continuous RL references become meaningful only after candidate support, masks, and oracle re-evaluation are trustworthy.

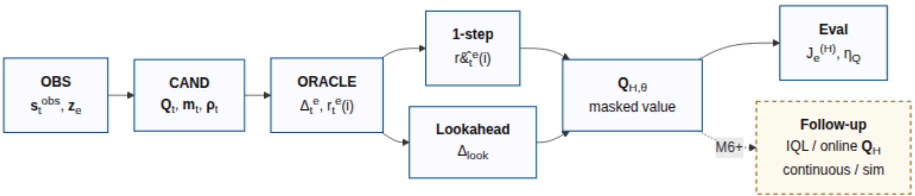

Figure 1: Compact evidence chain. The proposal moves from actor-visible s_t^{obs} and z_e through masked candidates, target RRI, lookahead headroom Δ_{look}, and masked Q_H; the dashed path is post-Q_H follow-up work.

4 Objectives and Hypotheses

The thesis objective is a leakage-safe target-aware NBV stack whose selected views reduce target point-mesh error under a fixed acquisition budget. The primary endpoint metric is J_e^H; the additive return G_t^H is the training target for value learning.

4.1 Aim 1: Target Oracle and Leakage Boundary

Define target-specific oracle RRI while keeping target selection and model input actor-visible. V1 uses observed or predicted target descriptors $\boldsymbol{z}_e$; GT crops, boxes, meshes, and all-candidate renders are restricted to labels, upper bounds, and evaluation. The required evidence is target eligibility, match score, unmatched/ ambiguous counts, endpoint J_e^H, and separate scene RRI and acquisition cost.

4.2 Aim 2: Target-Conditioned One-Step Control

Train a VIN-style myopic scorer over the same candidate table that later feeds Q_H. The scorer predicts target RRI from actor-visible scene, target, and candidate features. It is the required learned one-step control, evaluated by rank correlation, top-k oracle hit rate, calibration, selected-candidate oracle RRI, target visibility, invalid fraction, and grouped failures.

4.3 Aim 3: Non-Myopic Finite-Candidate Planning

First estimate whether bounded oracle lookahead has headroom over one-step oracle greedy:

$$\Delta_{\text{look}} = J_e^H(\pi_{\text{oracle-look}}) - J_e^H(\pi_{\text{oracle-1}}).$$

Only if this headroom is positive is Q_H expected to recover part of it from offline rollout traces. The candidate-query model emits one masked value per candidate:

$$\boldsymbol{u}_{t,i} = \text{Transformer}_{\theta(\boldsymbol{X}_t)_i},$$

$$Q_{H,\theta}(\boldsymbol{s}_t^{\text{cf0}}, \boldsymbol{z}_e, \boldsymbol{q}_{t,i}) = \boldsymbol{w}_Q^\top \boldsymbol{u}_{t,i},$$

$$a_t^\theta = \operatorname*{argmax}_{i \in \mathcal{A}_t} Q_{H,\theta}(\boldsymbol{s}_t^{\text{cf0}}, \boldsymbol{z}_e, \boldsymbol{q}_{t,i}).$$

Success is measured by oracle-rescored selected actions, not predicted values. If oracle lookahead itself has little headroom, the thesis reports that the current objective and candidate distribution are effectively myopic. If lookahead has headroom but Q_H fails to recover it, the analysis reports whether the limiting factor is target observability, candidate support, rollout coverage, reward definition, or model capacity.

Claim	Primary evidence	Decision rule
Target utility	J_e^H, G_t^H, scene RRI, cost	J_e^H decides endpoint quality; G_t^H trains and ranks rollouts.
Input safety	actor-visible $\boldsymbol{z}_e$, match score, support, leakage checks	GT is label/evaluation only for the V1 result.
Myopic control	target-rank metrics, selected-candidate oracle RRI, calibration	One-step target scoring is the comparator for Q_H, not the final policy claim.
Planning headroom	Δ_{look} and recovered fraction η_Q	Q_H is meaningful only relative to measured oracle-lookahead headroom.

Table 2: Objective-to-evidence matrix.

5 Method

5.1 Geometry, Targets, and One-Step Control

The geometry stage establishes the candidate poses, rig poses, rendered depths, backprojected point clouds, semi-dense support, EVL fields, and Rerun diagnostics that must agree in frame and indexing. Oracle labels use the implemented point-mesh terms

$$\Delta_t = \mathcal{A}_t + \mathcal{C}_t, \quad \Delta_t^e = \mathcal{A}_t^e + \mathcal{C}_t^e, \quad r_t^e = \frac{\Delta_t^e - \Delta_{t+1}^e}{\Delta_t^e + \varepsilon}.$$

The target stage adds the V1 contract. The selector returns a small top-K set of observed or predicted targets and refuses GT OBB input for the main result. Target labels are attached only after matching to GT crops. The one-step target-conditioned scorer keeps the current VIN-style ordinal setup, including CORAL variants [CMR19], and becomes the myopic learned control.

5.2 Candidate and Replay Contract

Each decision state carries a finite candidate table $\boldsymbol{Q}_t$, hard mask $\boldsymbol{m}_t$, invalid-reason vector $\boldsymbol{\rho}_t$, target descriptor $\boldsymbol{z}_e$, selected-view history, and remaining budget. Candidate generation is logged as a mixture over target-point, radial-shell, and forward-rig families. Each row stores provenance, pose, target/scene support, path increment, validity, reason code, and oracle labels. Candidate order has no semantics, so shuffled-candidate evaluation is required.

The minimum replay row is

$$\big(\text{scene}, \text{snippet}, \text{target}, t, \boldsymbol{s}_t^{\text{cf0}}, \boldsymbol{z}_e, \boldsymbol{Q}_t, \boldsymbol{m}_t, \boldsymbol{\rho}_t, a_t, r_t^e, \boldsymbol{s}_{t+1}^{\text{cf0}}, \boldsymbol{Q}_{t+1}, \boldsymbol{m}_{t+1}, \text{policy}, \text{seed}\big).$$

This row reproduces the mask, selected transition, value target, and oracle re-evaluation.

5.3 Rollout Policies and Q_H

The planning evaluation first compares deterministic one-step oracle greedy and bounded oracle lookahead:

$$\pi_{\text{oracle-1}(\boldsymbol{s}_t)} = \operatorname*{argmax}_{i \in \mathcal{A}_t} r_t^{e(i)},$$

$$\pi_{\text{oracle-look}(\boldsymbol{s}_t)} = \pi_0 \left(\operatorname*{argmax}_{a_{t:t+H-1}} \sum_{k=0}^{H-1} \gamma^k r_{t+k}^e \right).$$

Temperature-softmax traces are rollout-data diversity, not the final policy:

$$P(a_t = i \mid \boldsymbol{s}_t) = \frac{\exp\big(\beta \ell_{t,i}\big)}{\sum_{j \in \mathcal{A}_t} \exp\big(\beta \ell_{t,j}\big)}.$$

The learned planner is a candidate-query Transformer over scene, target, history, and candidate tokens [Lee+19, Vas+17, Zah+17]:

$$\boldsymbol{u}_{t,i} = \text{Transformer}_{\theta(\boldsymbol{X}_t)_i},$$

$$Q_{H,\theta}\big(\boldsymbol{s}_t^{\text{cf0}}, \boldsymbol{z}_e, \boldsymbol{q}_{t,i}\big) = \boldsymbol{w}_Q^\top \boldsymbol{u}_{t,i}.$$

Invalid candidates are filled with $-\infty$ before argmax, softmax, loss targets, and bootstrap maximization. With online parameters θ and target parameters θ^-, the first backup is masked Double-Q [HGS15, Mni+13]:

$$i^\star = \operatorname*{argmax}_{i \in \mathcal{A}_{t+1}} Q_{H,\theta}\big(\boldsymbol{s}_{t+1}^{\text{cf0}}, \boldsymbol{z}_e, \boldsymbol{q}_{t+1,i}\big),$$

$$y_t = r_t^e + \gamma(1 - d_t) Q_{H,\theta^-}\big(\boldsymbol{s}_{t+1}^{\text{cf0}}, \boldsymbol{z}_e, \boldsymbol{q}_{t+1,i^\star}\big),$$

$$\mathcal{L}_{Q(\theta)} = \frac{1}{|\mathcal{D}|} \sum_{(s,a,r,s') \in \mathcal{D}} m_{t,a} \big(Q_{H,\theta}\big(\boldsymbol{s}_t^{\text{cf0}}, \boldsymbol{z}_e, \boldsymbol{q}_{t,a}\big) - y_t\big)^2.$$

IQL, CQL, BCQ, sequence decoding, soft/energy policies, PPO, and SAC remain later comparison references unless the finite-candidate rollout store is stable [Che+21, FMP19, Haa+17, Haa+18, KNL21, Kum+20, Sch+17].

5.4 Evaluation

All selected actions are oracle-evaluated under the same acquisition and candidate budgets. The proposal uses one canonical comparison table:

Policy	Actor input?	GT decision?	Horizon	Primary role
π_{rand}	yes	no	1	lower reference over valid candidates
$\pi_{\text{learned-1}}$	yes	no	1	myopic learned target scorer
$\pi_{\text{oracle-1}}$	no	yes	1	one-step oracle upper bound
$\pi_{\text{oracle-look}}$	no	yes	H	non-myopic headroom estimate
π_Q	yes	no	H	learned recovery of lookahead headroom

Table 3: Policy comparison and leakage boundary. Report J_e^H, G_0^H, scene RRI, cost, invalidity, runtime, and coverage for each row.

6 Schedule and Risk Control

The planned thesis window runs from 29 April 2026 to 30 September 2026. The roadmap owns the detailed Gantt chart; the proposal keeps only milestone exit conditions.

Dates	Milestone	Exit condition
2026-04-29 to 2026-05-10	M0 proposal contract	Proposal, roadmap, research questions, and source policy state the same finite-candidate thesis claim.
2026-05-11 to 2026-05-31	M1 data/oracle	Offline-store, split, pose/frame, depth/backprojection, invalidity, Rerun, and throughput checks pass before target scale-up.
2026-06-01 to 2026-06-21	M2 one-step baseline	Scene-level VIN baseline, calibration plots, LRZ sharding plan, and Zarr rollout/Q schema are ready.
2026-06-22 to 2026-07-12	M3 target oracle	Target RRI, V1 OBS-SEL / PRED-Q / GT-EVAL, and observed-only target selection are trusted on a small subset.
2026-07-13 to 2026-08-09	M4 target scorer	Target-conditioned one-step scoring is compared with scene-level scoring and oracle target labels.
2026-08-10 to 2026-08-30	M5 headroom/Q_H	Oracle lookahead headroom is measured; Q_H is trained and oracle-evaluated if the headroom is positive.
2026-08-31 to 2026-09-13	M6 follow-up design	Online discrete Q_H, IQL, actor-critic, hierarchy, and simulator paths are written as follow-up designs or post-M5 ablations.
2026-09-14 to 2026-09-27	M7 experiments/writing	Final tables, figures, failure cases, coverage report, and thesis narrative are frozen.
2026-09-28 to 2026-09-30	M8 release	Configs, smoke checks, demo path, and final PDF artifacts are reproducible.

Table 4: Milestone exit criteria; the roadmap carries the full Gantt.

Three risks decide interpretation. If geometry or oracle labels fail M1, the thesis remains an oracle-contract and one-step-scoring study. If target matching is sparse or ambiguous, target RRI is reported only on validated subsets with explicit unmatched counts. If Δ_{look} is small, the M5 result is a falsifiable statement that the current candidate distribution is approximately myopic; continuous RL is not promoted as a substitute result.

7 Preliminary Thesis Outline

The thesis will be written as an empirical methods thesis. The chapter order mirrors the validation order so that claims are introduced only after the corresponding contract is established.

Chapter	Purpose	Expected evidence
Introduction	Motivate target-conditioned, quality-driven NBV for egocentric indoor reconstruction and state the finite-candidate thesis claim.	Research questions, contribution boundary, and source-backed scope.
Background	Review active perception, NBV, RRI, ASE/Project Aria, EFM3D/EVL, target-aware 3DGS, and offline value learning.	Literature synthesis with adoption/rejection decisions.
Data and geometry contracts	Describe snippets, calibration, frames, offline stores, candidates, rendered depths, backprojection, masks, and Rerun inspection.	Geometry contract report, visual diagnostics, throughput, and known limitations.
Oracle RRI and target RRI	Define scene/target distances, crop matching, invalidity, and label generation.	Label distributions, target crops, target/scene divergence, and failure cases.
Target-conditioned scoring	Present actor-visible target encoding, candidate features, VIN-style model, ordinal/regression losses, and calibration.	Held-out ranking, top-k oracle hit, ablations, calibration, and target-specific failures.
Bounded rollout and Q_H	Compare random-valid, one-step greedy, learned one-step scorer, oracle lookahead, temperature-softmax traces, and candidate-query Q_H.	Cumulative target RRI, endpoint target gain, scene RRI, cost, invalidity, runtime, and rollout visualizations.
Discussion and conclusion	Interpret limits, scale blockers, simulator gaps, semantic/global planning, and real-device follow-up paths.	Scope-bound conclusion and reproducibility package.

Table 5: Preliminary thesis chapter outline.

The expected final contribution is a reproducible target-aware finite-candidate view-selection study, not a broad claim that continuous reinforcement learning has been solved for egocentric reconstruction.

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